

RothCpy-AFM

**A Python version of the RothC model
accounting for different C-input sources
in agricultural and agroforestry systems
across diverse pedoclimatic conditions**

Reference Manual

Version 1

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The document contains the instructions on how to use the script and fill in the input data required to simulate soil organic carbon (SOC) dynamics according to the RothC methodology.

Installation and utilisation

A copy of the model (a Python script) and an input template (Excel spreadsheet) have uploaded to Zenodo, where the model is published. Additionally, a link to the GitHub repository dedicated to the RothCpy-AFM model is provided. Subsequent updates and modifications to the model will be disseminated via GitHub. A copy of this manual is also provided, containing instructions and suggestions on how to complete the required model inputs. The script can be run in Python [1], and Spyder is recommended as the most suitable IDE for scientific purposes.

Disclaimer

Accurate predictions rely on the understanding of the problems that characterise the context that the user intends to simulate. A calibration for the computation of the Carbon input entering the model workflow may be required, and validation against experimental data when those are available for the user case. In addition, knowledge of how to fill in the information for the different parameters based on the characteristics of your study area is of fundamental importance to correctly assess SOC dynamics. Therefore, the target users for this model should have a correct background on how to collect and pre-process information for the different input required by RothC computational procedures (e.g., deriving C-input from vegetation aboveground biomass).

License

This platform is designed for field-level applications and is intended primarily for academic and educational use. The original contents and components developed specifically for this platform, excluding third-party software, are made available for non-commercial use under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International licence (CC BY-NC-ND 4.0). Modified versions of these components may not be distributed without the prior written permission of the respective copyright holders.

The platform incorporates the Rothamsted Carbon Model (RothC), developed by Rothamsted Research and distributed under the Apache License 2.0, along with the modified version RothC10N (Farina et al., 2013). RothC remains subject exclusively to the terms of the Apache License 2.0, which permits its use, modification and redistribution, including for commercial purposes, subject to the conditions of that licence. Any modifications made to the original RothC code are identified as such and do not imply endorsement by Rothamsted Research.

Acronyms

AGB: aboveground biomass

BGB: belowground biomass

BIO: microbial biomass

DPM: decomposable plant material

FYM: farmyard manure

HUM: humified organic matter

IOM: inert organic matter

RPM: resistant plant material

SMD: soil moisture deficit

SOC: soil organic carbon

RothC and RothC10N

The Rothamsted Carbon model (RothC) originally developed by Coleman and Jenkinson (1996) (RothC-26.3) aimed to simulate C turnover in non-waterlogged topsoils. The model requires easily obtainable inputs, including: i) climatic variables (temperature, rainfall, and evaporation), ii) soil properties, mainly consisting of clay and initial soil organic carbon (SOC), and iii) C-inputs deriving from the agricultural management (crop residues and fertilization). Over the last decades, climate change has determined a variation of habitats in different parts of the world, with a gradual increase of arid and semi-arid areas. To address this, an early adaptation of the model (RothC26.5) was introduced for drylands [3]. However, subsequent studies demonstrated that both versions of RothC still struggled to accurately predict SOC dynamics in semiarid regions, particularly in rotations including fallow, often requiring unrealistic C-inputs to match field measurements (Lobe et al., 2005; Farina et al., 2013). This was mainly due to the simplifications to assess soil water retention under high evaporation conditions.

To overcome these limitations, a modified version of the model (RothC10N) was developed by Farina et al. (2013), taking into account a physically based approach to assess soil moisture deficit (SMD) and its effect on the decomposition of C-input entering the soil system. This approach considers additional soil hydraulic properties apart from the clay content alone, which is traditionally the only soil property used in standard RothC. The main differences between RothC and RothC10N regard: i) the minimum value of the rate modifying factor (RMF) for SMD, initially set to 0.2 in RothC, and lowered to 0.1 in RothC10N; ii) RMF for soil moisture deficit is calculated taking into account hydrological constants (field capacity, wilting point and hygroscopic water) determined from van Genuchten parameters [6], which are in turn estimated via the HYPRES pedotransfer function [7]. By modifying such parameters, soil decomposition rate reduces, improving model's performance in dry and semiarid regions [4].

This was used as the conceptual basis to implement the RothCpy-AFM model, considering the effect of C-input coming from different sources that are typically present in agroforestry (*i.e.*, an integrated land management combining woody perennial crops in agricultural systems). The version of RothC here presented also allows the user to simulate C turnover applying different methodologies, by choosing between the traditional version developed by Coleman and Jenkinson (1996), or RothC10N, which is the modified version developed by Farina et al. (2013), depending on the pedoclimatic conditions of the study area.

Required input and data completion

The input data required by RothCpy-AFM are quite simple and can be classified into two main groups: i) dynamic and ii) static data. The dynamic and static inputs are detailed below, with the required unit of measurement for each variable specified in the worksheet above the corresponding cell or column.

For each field on which the user wants to simulate SOC dynamics, both dynamic and static data must be entered into the Excel spreadsheet (Fig. 1). To simulate multiple fields, the user can copy-and-paste the template spreadsheet and populate it with the specific combination of inputs for every field that must be investigated. The user can assign any name they wish to the spreadsheet (e.g., the field ID), with the option to add abbreviations to indicate the features of each specific simulation. When naming an Excel spreadsheet, the user cannot use reserved characters such as slashes (/), colons (:), question marks (?), asterisks (*), or brackets ([,]). It is important that the user does not alter the cells containing the headings for the static (rows 3- 4) and dynamic (rows 6-7) inputs, as renaming them will prevent the script from recognising which inputs to assign during the simulation. Please ensure all numeric values use a period as the decimal separator, following the international format (e.g., 12.34 instead of 12,34), as the number of decimal places utilized has no effect on the underlying calculations.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	example input data to run RothC for a number of years													
2	clay and depth are only needed once, other data are monthly jan-dec													
3	(%)	(%)	(%)	(%)	(cm)	(%)	(t C/ha)	(0=N; 1=Y)	(t C/ha)	(0=Open-Pan-Evaporation, 1=EvapoTranspiration)	ratio of C-inputs during spin-up - Herbaceous	ratio of C-inputs during spin-up - Trees		
4	sand	silt	clay	skeleton	depth	Corg_pct	Corg_ton	IOMknown	IOM	ETknown	herb_share	tree_share		
5	29.31064	45.48716	25.2022	5	30	0.835154		0	nd	0	1	0		
6	(-)	(-)	(C)	(mm)	(mm)	(mm)	(t C/ha)	(t C/ha)	(t C/ha)	(-)	HERBACEOUS	TREE		
7	year	month	Temp	Prec	Irrigation	Evap	Herb_in	Tree_in	FYM	PC	CROP	OPERATION	YEAR	OPERATION
8	1	1	4.21	52.47	0	27.62	0	0	0	1				
9	1	2	5.58	71.08	0	54.41	0	0	0	1				
10	1	3	7.47	70.96	0	102.31	0	0	0	1				
11	1	4	10.77	77.52	0	142.69	0	0	0	1				
12	1	5	14.80	101.43	0	152.27	0	0	0	1				
13	1	6	20.32	49.27	0	213.81	0	0	0	1				
14	1	7	23.54	34.44	0	239.91	0	0	0	0				
15	1	8	23.36	44.29	0	214.22	0	0	0	0				
16	1	9	18.64	63.19	0	135.84	0	0	0	0				
17	1	10	14.39	86.29	0	75.28	0	0	0	0				
18	1	11	9.02	92.58	0	40.88	0	0	0	0				
19	1	12	5.63	94.83	0	21.65	0	0	0	0				
20	2018	1	6.08	46.84	0	32.49	0	0	0	1				
21	2018	2	1.08	208.24	0	37.05	0	0	0	1				
22	2018	3	5.52	156.80	0	75.53	0	0	0	1				
23	2018	4	13.79	49.11	0	160.34	0	0	0	1				

Fig.1 Example of compiled Excel spreadsheet for both dynamic and static inputs required by RothCpy-AFM. For the dynamic data, columns K:N (rows 6-7) are optional data, that can be compiled by the user to keep track of the information stored in the previous columns for the execution of the script. On the other hand, values for columns A:L regarding the static input data must all be completed (apart some exceptions that are indicated in the following paragraphs).

Dynamic input data

Dynamic data (Table 1) must be provided on a monthly time-step, where each entry is explicitly identified by the corresponding calendar year and month in the Excel spreadsheet. These data consist of the following variables:

- Climate data:
 - o Monthly average temperature (Temp);
 - o Monthly cumulated rainfall (Prec);
 - o Monthly cumulated irrigation (Irrigation): in other versions of RothC, this value was included in the cumulated rainfall. This version of the model keeps the two distinct inputs, and the script performs the sum of these values;

- Monthly evaporation (Evap): in the original version, this parameter corresponded to the open-pan evaporation, which was then converted into evapotranspiration using a coefficient. To date, since potential evapotranspiration is most widely used and is easier to obtain, we added a binary condition statement (ETknown) in RothCpy-AFM allowing the script to switch between computation formulas, depending on the data available for a given area that the user plans to simulate.

NOTE: the script switches formulas depending on the ETknown condition statement, make sure that the values provided and the condition statement are coherent.

- Farm management data:

- C-input: this version of RothC was developed to simulate agroforestry conditions, in which both herbaceous and woody crop residues can be simultaneously present in the field. The model keeps these inputs (Herb_in and Tree_in) separate to account for the different crop residues and lignin content of herbaceous and tree species, applying a distinct DPM/RPM ratio to calculate the decomposition of each material. This separation also allows to account for the presence of vegetation in conventional plantation forests, where spontaneous or controlled herbaceous vegetation can be present in the interrow.
- Farmyard manure (FYM): in this category are included monthly inputs coming from organic fertilizers of any kind. Those are treated separately from crop residues because are assumed to be more decomposable than normal plant material [8]. Mineral fertilizers are not contemplated in the RothC computational procedures.
- Plant Cover (PC): refers to the presence of active vegetation on the soil surface. PC (in other works also indicated as soil cover) is a binary variable (0-1) that respectively takes the value of 1 in case of bare soil, and 0.6 otherwise. This lower value slows down the decomposition rate. Combined with temperature and SMD, it determines the monthly RMF for decomposition, while also affecting evapotranspiration limits.

Table 1. Dynamic parameters required for the RothCpy-AFM model.

Parameter	Description	Unit	Default value
Temp	Average temperature.	°C	-
Prec	Cumulated rainfall.	mm	-
Irrigation	Monthly irrigation volumes	mm	-
Evap	The user can choose whether to use the open-pan evaporation or the evapotranspiration. This parameter is also linked to the correct assignment of the condition ETknown.	mm	-
Herb_in	C-input entering the soil, derived from herbaceous crops residues.	MgC ha ⁻¹	-
Tree_in	C-input entering the soil, derived from tree crops residues.	MgC ha ⁻¹	-
FYM	C-input resulting from agronomic management (manure and other organic soil amendments).	MgC ha ⁻¹	-
PC	Plant cover.	0 = no cover, 1 = plant cover	-

Farm management inputs regarding plant carbon coming from crop residues, applications of organic fertilizers and amendments (FYM) and periods of bare or vegetated soil (plant cover) reflect the actual observed crop rotations, changes in land use or management practices that one wishes to evaluate. This module enables the quantification of the impact of climate change or different agronomic strategies on carbon storage in the medium and long term.

A focus on the quantification of monthly plant carbon input required by the model is one of the most critical variables in this part of the simulation. Such category includes not only aboveground biomass (AGB) crop residues (stubble, straws, stalks, leaves, etc.), but also the belowground biomass (BGB, mainly consisting of

plant's root system) and roots' exudates, along with any weeds. Determination of C-inputs required by RothC can follow different approaches.

Whenever possible, C-inputs should be derived through laboratory analysis of sampled plant residues. Despite the high accuracy of this approach, it can also be extremely time-consuming, destructive and costly, making them difficult to apply on a large scale or for retrospective historical simulations.

Another approach is the indirect assessment. In standard modelling, it is often more efficient and sustainable to apply standardised methods to derive C-inputs from readily available field data (both measured or derived from open-source and scientific datasets). Starting from either the grain yield or the AGB, depending on the specific crop or data availability, parameters such as the Harvest Index (HI) for arable crops, and specific root-to-shoot ratios (R:S) can be used to mathematically reconstruct plant's total biomass (AGB+BGB) and its subsequent carbon content via allometric functions for both tree and herbaceous crops. The main scientifically recognised methodologies include international guidelines and national inventories [9–11], prediction models that divide plant carbon in different fractions, based on empirical approaches [12–15], or mechanistic models simulating the actual biophysical and physiological processes of the crop in response to climate and management practice [16]. However, the actual C returning to the soil relies not only on total biomass production, but strongly depends on crop destination (e.g., grain vs. forage) and post-harvest residue management (retained or removed). Forage systems yield minimal soil C-inputs, limited mainly to roots, exudates, and weeds, whereas grain crops with retained residues provide significantly higher organic carbon inputs.

Static input data

Static data (Table 2) include properties of the topsoil (depth set by default to 30 cm). These inputs are necessary to model initialisation and computational procedures. In addition to the previous versions of the model, a couple of flag conditions are included, describing the type of certain input data that are fed to the model.

- Soil properties:
 - o sand, silt, and clay percentages: in particular, clay influences soil's water holding capacity and the fate of decomposed carbon between the pools composing SOC (DPM, RPM, BIO, HUM and IOM) and is used to calculate the RMF for soil moisture deficit. High clay contents reduce the proportion of C released into the atmosphere as CO₂, retaining a greater proportion in the soil in the form of microbial biomass (BIO) and humus (HUM). If the user chooses to run the RothC10N model, values of the other mineral soil fractions (sand, silt) are required to calculate soil hydrological properties and assess the RMF for soil moisture.
 - o skeleton percentage in volume: when the total initial C_{stock} is unknown, this parameter enters the formula used in RothCpy-AFM to convert the initial soil organic carbon content (C, expressed in %) into the total organic carbon stock (SOC_{stock}, in Mg C ha⁻¹). This conversion accounts for both soil depth and bulk density (BD), the latter being estimated using the pedotransfer function proposed by Rawls et al. (1982) (see Equation 1 in [17])
 - o Soil depth: by default, RothC simulates SOC dynamics in the upper soil layer (0-30 cm depth).
 - o Corg_pct: soil organic C concentration (as percentage) as routinely provided by standard soil laboratory analyses. Together with the previously described soil properties, it can be used to assess the Soil Organic Carbon stock (SOC_{stock}, expressed as MgC ha⁻¹) within the script. In this case, the user can bypass the insertion of the following input value.
 - o Corg_ton: is the Soil Organic Carbon stock (SOC_{stock}, expressed as MgC ha⁻¹) within the simulated soil layer at time zero. Since both RothC and RothC10N simulate C dynamics as carbon mass per unit area across the five pools (DPM, RPM, BIO, HUM, IOM), this value serves as the target

benchmark for the spin-up phase. If directly measured stock data are available, the user can provide `Corg_ton` and bypass `Corg_pct`.

- Inert organic matter (IOM): represents the chemically protected fraction of SOC that does not participate in decomposition cycles and remains constant over time. During the initialization phase, IOM can be provided as an estimated or measured value. Conversely, the model can automatically derive it solely from the initial SOC data at time zero, using the standard Falloon equation [18].
- Flag conditions specifying the type of data (`IOMknown` and `ETknown`, respectively). The correct assignment of these flag variables is fundamental to avoid errors in simulating SOC dynamics.
- Breakdown between carbon inputs coming from herbaceous (`herb_fraction`) and tree (`tree_fraction`) components. These values are used to achieve equilibrium during the spin-up phase. The sum of these two values must not exceed 1.

Table 2. Fixed parameters required for the RothCpy-AFM model.

Parameter	Description	Unit	Default value
<code>sand</code>	Soil sand content.	%	-
<code>silt</code>	Soil silt content.	%	-
<code>clay</code>	Soil clay content.	%	-
<code>skeleton</code>	Soil skeleton content.	%	0
<code>depth</code>	Soil layer depth.	cm	30
<code>Corg_pct</code>	Organic carbon. If provided, you can bypass <code>Corg_ton</code>	%	-
<code>Corg_ton</code>	Soil Organic Carbon stock in the analysed layer If provided, you can bypass <code>Corg_pct</code>	MgC ha ⁻¹	-
<code>IOMknown</code>	Flag to assess if IOM is known by the user, and has to be typed in the subsequent cell of the spreadsheet	0 = unknown 1 = known	-
<code>IOM</code>	Inert organic matter content of the simulated soil. If unknown, it must be specified in the the <code>IOMknown</code> cell, and the model will automatically compute it from the <code>Corg_pct</code> or the <code>Corg_ton</code> variables	MgC ha ⁻¹	Leave blank if value is unknown
<code>ETknown</code>	Flag to specify the type of climate evaporation data used for the simulation.	0 = open-pan evaporation 1 = evapotranspiration	-
<code>herb_fraction</code>	Fraction indicating the weight of C-inputs coming from herbaceous residues	-	0-1 range
<code>tree_fraction</code>	Fraction indicating the weight of C-inputs coming from tree residues	-	0-1 range

Running the script

Upon launching the script, a message appears in the Python IDE console, prompting the user to select the model (RothC or RothC10N) deemed most suitable for his specific case study. Neither version of the biodynamic model is appropriate for environments prone to waterlogging. If the study area is in a semi-arid zone or characterized by challenging climatic conditions and water availability (e.g., rain-fed crops), RothC10N is preferable for simulating SOC dynamics; otherwise, the original version (RothC) may be used.

Spin-up simulation

This phase of the simulation (also called equilibrium or inverse mode) aims to reconstruct an accurate scenario of the historical SOC dynamics for the simulated field before time zero via an iterative calibration cycle. To do so, the model requires a representative standard year built with average climate data (temperature, rainfall and evaporation) reflecting the climatological trends for the investigated area. Typically, a good climatological timeseries is obtained using at least 30 years of measurements. Dynamic input data also include the average field management, consisting of plant cover status, land cover type (forest/tree crop/fallow/arable crops) and the corresponding C-inputs, which must be configured by the user according to the following guidelines:

- PC: is set to 1 when vegetation is active, and 0 when there is no active vegetation, or the plant enters a dormancy stage;
- Monthly C-inputs from crop residues: the user must leave the target cells empty in the Herb_in and Tree_in columns for the months when carbon inputs are expected to occur (e.g., at harvest or tillage).
- The user must assess whether the carbon inputs during this phase come from only the herbaceous or tree component or has to be partitioned between these two sources by assigning a value to the variables herb_fraction and tree_fraction.

Instead of requiring a manual adjustment of a hypothetical plant carbon input entering the soil during the spin-up phase, this section of the RothCpy-AFM script automates the optimization process to determine the annual carbon input required to match the simulated SOC value with the baseline SOC measured at time zero. This is how code calibration logic works:

- 1) The routine detects the empty cells within the 12-month equilibrium dataset for both Herb_in and Tree_in columns. These cells serve as placeholders, instructing the script where to allocate the carbon input quantities required by the model to reach a steady state.
- 2) To efficiently identify the required amount of C-input for the spin-up phase, a bisection algorithm is employed. Starting from a flexible, user-defined interval (bounded by the low_c and high_c variables) accounting for diverse soil productivity levels and managements (Fig. 2), the algorithm systematically narrows down the candidate C-input value within a range, then the final value is assigned across the uncompiled months detected in the equilibrium dataset. If missing cells are detected in both herbaceous and tree input columns, the input is split according to the fractions (herb_fraction and tree_fraction) specified in the static input data (Table 2).
- 3) For each C-input candidate, the model iteratively runs the spin-up module until when active soil pools (DPM, RPM, BIO, HUM) reach a steady state for a given carbon input amount. That is achieved evaluating a pool-by-pool convergence between values at the beginning and the end of every iteration.
- 4) The routine terminates once the difference between modelled SOC and the baseline SOC provided by the user falls below a convergence tolerance threshold, successfully identifying the equilibrium carbon input.

```

932 # COMPUTE C-inputs pre-spinup phase
933 df_eq, optimal_annual_c = optimize_c_inputs_for_spinup(df_eq_template=df_eq,
934 df_head=df_head,
935 Corg_target=Corg_init,
936 IOMinit=IOMinit,
937 tol = 0.5, #!!! eventually increase/decrease value
938 low_c = 0.5, #!!! adjust accordingly to your case study if necessary
939 high_c = 8.0, #!!! adjust accordingly to your case study if necessary
940 max_iter = 10, #!!! if low_c & high_c range is small, iterations could be reduced
941 herb_pct = herb_fraction,
942 tree_pct = tree_fraction,
943 sheet_name=str(sheet))
944

```

Fig. 2 Section of the RothCpy-AFM script. The lines with the checkmark icons indicate the variables that can be modified to calculate C-inputs (from either herbaceous crops or tree residues) required to complete the spin-up phase.

Spin-up simulations are always performed applying the standard RothC model. During this phase, extreme accuracy in assigning monthly timing and precise residue splitting to achieve equilibrium is not a critical issue. However, the partitioning of C-inputs between herbaceous and tree residues must ensure ecological coherence with the primary historical land cover (e.g., attributing baseline C-inputs to tree residues, in our case, provided historical poplar plantations rather than simulating a long-term arable crop sequence).

Once the spin-up phase is completed, the simulated values for the five soil carbon pools are used as initial conditions to launch the actual simulation, referred to as the Forward simulation.

Forward simulation

Unlike the spin-up phase, this simulation describes actual site-specific soil dynamics with high-resolution, real-world monthly datasets regarding climate data, alongside the most accurate available data for soil properties and farm management. Actual weather time series (historical or future climate scenarios) specific to each year of the simulated period are used.

For a more practical execution of multiple simulations, the excel input file template can be replicated as many times as necessary if the user must perform multiple simulations (i.e., simulations for multiple fields and/or scenarios). The use of a scriptable version of RothC allows to automate the passage from one simulation to the other, resulting in a reduction of the times required to run the simulation once the input data are correctly filled into the excel worksheet.

Output data

The output provided by the model consist of a series of .CSV and .TXT files for every simulation that is performed:

- Monthly output data:
 - o “filename_out_init_vals_until_eq.txt”: stores the results of every simulation performed during the spin-up, to keep track of how the soil carbon pools balance until reaching equilibrium;
 - o “filename_out_DF_MgC_ha-1_RAW_data.csv”: contains all the intermediate steps performed by RothCpy-AFM to obtain the final carbon pools and SOC_{stock} (monthly data);
 - o “filename_out_DF_MgC_ha-1_simplified.csv”: this file compiles the key model information (i.e., the monthly partition between DPM, RPM, BIO, HUM, IOM, CO₂ and SOC_{stock})
- Yearly output data:
 - o “filename_out_DF_MgC_ha-1_yearly.csv”: contains the same information as the .CSV file before, but data are yearly aggregated.

The script also includes a series of graphs providing an immediate trend for the simulations, but the user can feel free to graphically edit the data that are stored inside the aforementioned .CSV files

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